A Detailed Comparison of Two WAAM Simulation Codes

Original *WAAM-MultiLayer-Simulation* (FreeFEM++ single script) versus the derived *WAAM-Framework* (multiphysics FEM platform)

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Scope of this document. Two code bases held under `Desktop/WAAM/` were compared line by line: (i) the *original* upstream repository `WAAM-Framework/original_git_file/` containing `README.md` (434 lines) and `downloaded.edp` (534 lines) — a single hand-written FreeFEM++ script; and (ii) the *framework* `WAAM-Framework/`, containing 28,595 lines of Python across 40 source files, 3,571 lines of reference `.edp` modules, and a code generator that emits scripts of up to ∼3,150 lines. All line counts, file names and line references below were measured from the working tree on 13 October 2026.

1. Executive Summary

Table 1: Head-to-head summary of the two code bases.

Dimension	Original	Framework
Nature	One hand-written FreeFEM++ script plus a README	Layered Python application that *generates* FreeFEM++ scripts
Scale	968 lines, 2 files	28,595 lines Python + 3,571 lines `.edp` + generator (∼40×)
Workflow	Edit `.edp` → `FreeFem++ waam_multilayer.edp` → open `waam.pvd` in ParaView	GUI (12 tabs) or CLI → slice STL → mesh → generate → solve (serial/MPI) → plots + HTML report
Geometry	Hard-coded 26 × 3 mm wall on a 200 × 100 × 10 mm plate	Arbitrary STL/STEP/IGES/BREP, parametric walls, G-code-driven toolpaths
Heat source	Goldak double ellipsoid only	7 selectable models
Material	Mild steel S235 hard-coded	8-alloy JSON database, dual substrate/deposit assignment
Mechanics	Single post-hoc *elastic* solve, inherent strain	Incremental **J2 thermo-elasto-plasticity**, Newton with consistent tangent, TRIP, springback
Thermal extras	Conduction + convection	+ radiation, boiling cap, enthalpy-exact latent heat, adaptive Δt, Picard correctors, target-temperature inter-pass control
Beyond original scope	—	OpenFOAM VOF melt-pool CFD, ultrasonic (UVA) assistance, microstructure and Vickers hardness prediction
Verification	None	11-group physics suite, J2 cross-check, external CAS-DED benchmark, pytest suite
Portability	`D:\freefem++\...` hard-coded	Binary discovery + environment overrides, MPI, self-contained timestamped run directories

One-sentence verdict. The original is a *research artifact* — correct, minimal and self-documenting — while the framework is an *engineering platform* that retains the original physics as its inner core and rebuilds everything around it: geometry, toolpath, materials, plasticity, user interface, verification and reporting.

2. The Original: What It Actually Does

2.1 Physics implemented in `downloaded.edp`

- **3D transient heat equation**, θ-scheme with $\theta = 0.5$ (Crank–Nicolson), P1 elements, UMFPACK direct solver (lines 196–225).
- **Goldak double-ellipsoid** moving source with $a_f = 8$, $a_r = 20$, $b_G = 7$, $c_G = 4$ mm, $f_f = 0.6$, $f_r = 1.4$, a tanh front/rear split ($\varepsilon_{\text{split}} = 10^{-3}$) and amplitude

$$Q_0 = \frac{6\sqrt{3}\, f P}{\pi\sqrt{\pi}\, a b c}, \qquad P = I U \eta = 200 \times 24 \times 0.8 = 3840 \text{ W}. \tag{1}$$

- **Process schedule**: $v_{\text{scan}} = 10$ mm/s, $t_{\text{pass}} = 20$ s, $t_{\text{wait}} = 60$ s, $t_{\text{total}} = 2080$ s, 26 layers.
- **Quiet-element activation**: inactive deposit nodes receive $k_{\text{eff}} = 10^{-4}k$ and $C_{p,\text{eff}} = 10^{-4}C_p$; nodes are born progressively when $x_{\text{torch}} \geq x_{\text{node}}$ inside the weld loop; the source is *masked* by ψ_{active} so the arc cannot heat material that does not yet exist (lines 271–309). The README identifies this as the number-one correctness requirement: without it, $C_p \approx 0$ produces 10^7 °C spikes.
- **Temperature-dependent properties**: $k(T)$ falling 60 → 38 → 27 W/mK, $C_p(T)$ with a Curie peak of 5000 J/kgK at 720 °C, $T_{\text{sol}} = 1450$ °C, $T_{\text{liq}} = 1520$ °C.
- **Boundaries**: $h_{\text{bot}} = 50$ on face 1 (bottom), $h_{\text{conv}} = 15$ on faces 2–6. *No radiation, no fusion latent heat, no boiling cap* — conduction and convection only.
- **Fixed three-step timestepping**: $\Delta t_{\text{weld}} = 0.333$ s (so $v\Delta t \approx 0.5\, h_x$), $\Delta t_{\text{wait}} = 10$ s, $\Delta t_{\text{cool}} = 120$ s × 20 steps.
- **Mechanical stage (post-hoc)**: accumulate eigenstrain $\varepsilon^{\text{th}} \mathrel{+}= \alpha\, \Delta T$ with $\alpha = 2.1 \times 10^{-5}$ at every step, then a single linear-elastic solve with $E(T)$ from 210 to 80 GPa and $\nu = 0.3$, bottom face fixed, written to `waam_mech.vtu` (lines 464–525).

2.2 Mesh and output

Two `cube()` meshes are concatenated with `Th = ThBase + ThDep`: base $30 \times 20 \times 8$, deposit $30 \times 6 \times 26$, scaled by linear `movemesh3` only (a documented FreeFEM 4.15 / Windows workaround forbidding trigonometric transforms). The result is approximately 11,687 nodes and 56,880 tetrahedra, with one element per layer in z.

Outputs are `waam.pvd`, `T_NNNN.vtu` (fields `TempC`, `Activation`), `waam_mech.vtu` and `waam_data.csv` with eight columns, all written to the hard-coded path `D:\freefem++\arc_waam_simulation\`.

2.3 Known quirks in the original

1. **Header/code mismatch.** The comments promise "activation via a gradual tanh ramp over 5 s" and "$T = T_{\text{liq}}$ on activation"; the code switches ψ_{active} instantly at $x_{\text{torch}} + a_f$ and initialises at T_{amb} (deliberately — see the comment at line 92 warning that $T_{\text{dep,init}} = T_{\text{liq}}$ causes double heating).
2. **Non-conforming interface.** `ThBase + ThDep` merges no nodes: the substrate has 5.0 mm y-spacing and the deposit 1.33 mm, so the two blocks are not bonded.
3. **No radiation / latent heat / evaporation** despite peak temperatures near 1900 °C.
4. **Elastic-only residual stress** — no yield limit, so the reported von Mises stress can exceed the material's strength.

3. Architecture: Monolith to Platform

The structural change that dominates every other difference is that *the physics stopped being the source code and became an emission target*. `EDPBuilder.build()` (`codegen/edp_builder.py`, line 1474) assembles a script from more than twenty ordered sections:

```
GEOMETRY → PROCESS → FEEDSTOCK → MATERIAL → TIMESTEP → G-CODE READER → MESH → ADAPTIVE dt
→ FE SPACES → THERMO-ELASTO-PLASTIC ENGINE → FORMS → PROBES → OUTPUT → RESTART → DEPOSITION
LOOP → FINAL COOLING → MECHANICAL → MICROSTRUCTURE → VALIDATION GATE → solver_results.json
```

Table 2: Largest source files by line count (excluding caches and run artefacts).

Lines	File	Role
6 959	`gui/app.py`	Tkinter desktop GUI, 12 tabs
4 992	`codegen/edp_builder.py`	FreeFEM script generator
1 958	`core/fluid/openfoam_coupling.py`	OpenFOAM case generation
1 464	`tests/physics_suite.py`	279 automated checks
1 419	`postprocess/plot_generator.py`	8 publication plots
1 331	`gui/widgets/waam_3d_studio.py`	Three.js WebGL viewer
1 255	`core/postprocess/output_manager.py`	Summaries, reports
1 242	`gui/widgets/mesh_viewer_3d.py`	Embedded PyVista viewport
1 174	`postprocess/report_generator.py`	HTML engineering report

Original: `downloaded.edp` = 534 lines, `README.md` = 434 lines

Two hand-maintained `.edp` files survive as static presets rather than generator output:

- `DEFAULT.edp` / `generated_edps/DEFAULT.edp` (655 lines) — the original's architecture *with fixes*: an enthalpy-exact piecewise-linear C_p band over $[T_{\text{sol}}, T_{\text{liq}}]$, a quiet method that quenches only k (retaining full mass so diffusivity collapses instead of thermal mass), linearised Stefan–Boltzmann radiation ($\varepsilon = 0.7$), a boiling cap at $T_{\text{boil}} = 2862\,^{\circ}\text{C}$ with an E_{cap} energy audit, and an explicit caveat that the mechanical stage is a "post-hoc ELASTIC estimate (no plasticity)";
- `generated_edps/uva_waam_ss308l_ss316.edp` (859 lines) — SS316 substrate with SS308L deposit over 33 layers and *ultrasonic vibration assistance* (20 kHz, 15 μm, Blaha yield softening).

4. Geometry and Meshing

Table 3: Geometry and mesh pipeline comparison.

	Original	Framework
Input	None (hard-coded)	STL, STEP, STP, IGES, IGS, BREP, PLY, OBJ, G-code
Mesher	FreeFEM `cube()` $\times 2$	Gmsh OCCT volume meshing (`mesh/stl_pipeline.py:541`) *or* pure-Python graded structured mesh (`mesh/structured_conforming.py:464`)
Sizing	Fixed $30 \times 20 \times 8$ / $30 \times 6 \times 26$	Rule-based engine, 3 strategies, node budget 1500/2500/5000/custom (`mesh/intelligent_mesher.py:29`)
Interface	`ThBase + ThDep` → **zero-width crack**	**Single conforming mesh, nodes shared by construction**
Labels	Implicit	Enforced: 1 = bottom, 5 = top, 2/3/4/6 = flanks, shared by mesher and generated forms
Quality control	None	Signed-volume, orientation and outward-normal checks; the mesher *raises* rather than emit an unusable mesh (`edp_builder.py:3168`)

The conforming-mesh work is the framework's most consequential *correctness* fix. The crack produced by the original's mesh union was measured on a running model at a **19.9 °C mean and 92.9 °C peak** temperature discontinuity across the interface — i.e. no genuine conduction and no mechanical bond between substrate and deposit.

5. Toolpath: Linear Scan to Slicer and G-code

- **Original:** the torch traverses $x = 0 \rightarrow 200$ mm in one direction, every layer, at constant v_{scan}.

- **Framework:** `mesh/slicer_api.py` (695 lines) slices geometry into layers using five infill patterns (zigzag, meander, contour, crosshatch, unidirectional) with overlap, perimeters and bidirectional reversal, emitting `toolpath.gcode`, `toolpath.txt` and `layer_times.txt`. `mesh/gcode_parser.py` (411 lines) parses G0/G1/**G2/G3 arcs** into solver arrays.
- The generated script reads `toolpath.txt` (t, x, y, z, power) and `layer_pass_times.txt`, gates the arc off during rapid travels, supports a direction sign for zig-zag layers, and clamps the activation lookup with an "afterProgram" guard so an over-energised arc is not re-applied after the program ends.

6. Thermal Physics

Table 4: Thermal formulation comparison.

Aspect	Original	Framework
Scheme	$\theta = 0.5$, one solve per step	θ-scheme + Picard correctors (3 passes, 0.5 °C tolerance) for non-linearity
Timestepping	Three fixed steps (0.333 / 10 / 120 s)	Fixed *or* adaptive Fourier-number controller (Fo target 0.45, torch-advance cap, runaway guard)
Latent heat	C_p spike to 5000 J/kgK at the Curie point only; no fusion latent	Enthalpy-exact piecewise-linear C_p band over $[T_{\mathrm{sol}}, T_{\mathrm{liq}}]$ preserving $\int C_p\,dT$ (fixes ∼10 % energy error)
Radiation	None	Linearised Stefan–Boltzmann on all outer surfaces, emissivity per alloy
Evaporation	None	Boiling cap at T_{boil} with E_{cap} energy accounting
Inter-pass control	Fixed $t_{\mathrm{wait}} = 60$ s	`time` mode (as original) *or* `target` mode — iterative cooldown to a target peak temperature (default 400 °C) with convergence test and iteration cap
Quiet elements	k and C_p both scaled by 10^{-4}	k quenched with full mass retained; plus optional prescribed-T birth (Ansys-style liquidus gate)
Energy audit	`Tmax` printed to the console	`solver_results.json` balances intended arc, injected arc, stored enthalpy, boiling-cap removal, surface losses, quiet-node discard, floor additions and numerical residual

7. Heat Sources

Table 5: Available volumetric and surface heat-source models.

Model	Orig.	Fwk.	Notes
Goldak double ellipsoid	**Yes**	**Yes**	Identical algebra to Eq. (1) — direct carry-over
Gaussian volumetric	**No**	**Yes**	$Q_0 = 3\sqrt{3}P/(\pi\sqrt{\pi}R^3)$
Gaussian surface	**No**	**Yes**	Neumann `int2d` boundary condition
Conical frustum (keyhole)	**No**	**Yes**	$R(\zeta) = r_t + (r_b - r_t)\zeta/d_{\mathrm{pen}}$
Uniform band	**No**	**Yes**	tanh-smoothed box
Prescribed temperature	**No**	**Yes**	Post-solve $T = T_{\mathrm{liq}}$ inside the activation radius
P48 arc/droplet partition	**No**	**Yes**	*Prog. Addit. Manuf.* 1 (2025) 10:11647; empirical $\sigma_q = 6.205 \times 10^{-4} I^{0.28}$

All models remain masked by ψ_{active} — the original's key insight, preserved intact. Automatic source sizing keeps the semi-axes at or above the local element size with a fidelity floor (`_auto_size_source`, `edp_builder.py:468`). The six files under `core/heat_sources/*.edp` are explicitly marked *standalone reference modules — not wired into the pipeline*; the live physics is whatever `EDPBuilder` emits.

Table 6: Material data comparison.

	Original	**Framework**
Source	`kSteel(T)`, `CpSteel(T)` written inline	`core/multimaterial/material_database.json` (1,309 lines)
Alloys	1 (S235/A36 mild steel)	8: AlSi10Mg, ER70S-6, IN625, Pure Al, SS304, SS308L, SS316L, Ti-6Al-4V
Data	k, C_p, ρ, T_{sol}, T_{liq}, E, ν	+ ν, latent heat, emissivity, composition, temperature tables for $k/E/\sigma_y/\alpha$, phase transformation, hardness coefficients, wire diameters
Assignment	One material for the whole domain	**Dual**: substrate and deposit alloys chosen independently, with a dissimilar-metal CTE/thermophysical compatibility report
Phase kinetics	None	`core/multimaterial/alloy_metallurgy.py`: 5 phase families $\times$ 4 phases (Burgers BOR, WRC-1992 + Hall-Petch, Scheil/Laves, cellular eutectic, Kirkaldy-Venugopalan + Koistinen-Marburger), IIW carbon equivalent, WRC Cr/Ni equivalents

8. Materials

9. Mechanics: Elastic Estimate to Incremental Plasticity

Table 7: Mechanical formulation comparison.

	Original	**Framework**
Formulation	One post-hoc linear-elastic solve	Incremental J2 thermo-elasto-plasticity, backward-Euler radial return
Tangent	n/a	Consistent-tangent Newton (default) or Picard, with force and displacement residuals recorded
Hardening	None	Isotropic H'
TRIP	None	Transformation strain $\beta = 0.011$
Remelt	Eigenstrain keeps growing unphysically	Plastic memory reset on remelt; above-solidus "slurry" stiffness cut avoids singular degrees of freedom
Eigenstrain	ε^{th} accumulated from $t = 0$, can exceed yield $\rightarrow$ phantom stress	Correct eigenstress subtraction; `DEFAULT.edp` documents the phantom-stress fix
Boundary conditions	Bottom fixed only	Isostatic / fixture-clamped / cantilever, fixture stiffness, springback with fixture-release regularisation, gravity
Discretisation	P1 only	P1 or P2; legacy inherent-strain fallback retained but disabled
Output	`waam_mech.vtu`, maxima to console	+ PEEQ, per-frame von Mises, Newton convergence in `solver_results.json`, checkpoint/restart

10. New Physics Domains Absent from the Original

1. **OpenFOAM melt-pool CFD.** `core/fluid/openfoam_coupling.py` (1,958 lines) generates an OpenFOAM 2412 `beamWeldFoam` case: volume-of-fluid two-phase (metal plus atmosphere), diverging-oscillating-Gaussian melting source, Marangoni shear, vapour recoil and a Darcy mushy-zone sink. It maps $IU\,\eta \rightarrow$ `MELTING/HS_Q`, $v_{\text{scan}} \rightarrow$ `HS_velocity` and $b_G \rightarrow$ `HS_a`, then writes `cfd_history.csv`, plots and `cfd_report.html`.

2. **Ultrasonic vibration assistance (UVA).** 20 kHz / 15 μm with four acoustic modules (streaming, heat generation, cavitation, momentum transfer); amplitude acts on the *yield surface* (Blaha softening) rather than on Young's modulus, with a layer-by-layer multi-material zone table.
3. **Microstructure and hardness.** G/R solidification map, JMAK kinetics, secondary dendrite-arm-spacing laws per alloy family, reheat anneal factors and discrete Vickers indentation predictions, producing `hardness_profile_z.csv` and `hardness_test_locations.csv`.
4. **Reference modules not yet wired:** Marangoni melt pool, vibration acoustics and solidification-phase `.edp` files under `core/`; the `core/thermal/` and `core/mechanical/` directories are empty.

11. Output and Postprocessing

Table 8: Output artefacts and postprocessing.

	Original	**Framework**
Fields	`TempC`, `Activation`, `Ux Uy Uz DispMag SigMises`	+ PEEQ, per-frame mechanics, UVA fields, `VickersHV`
Files	`waam.pvd`, `T_*_*.vtu`, `waam_mech.vtu`, `waam_data.csv`	+ `solver_results.json`, `run_summary.json`, `parameters.json`, `probe_points.csv`, `probe_paths.csv`, hex-VTK conversions, `waam_3d_studio.html`
Plots	None — the README asks the user to write matplotlib by hand	8 publication figures at 300 dpi: thermal history and peaks, cooling rates and $t_{8/5}$, distortion and warpage, von Mises, heat input and dwell energy, baseplate thermal cycle with elevation probes, spatial stress paths (sections A–A, B–B), normalised warpage index
Reports	None	`report.html` (two-page A4), `run_summary.json`, ParaView loader script, `cfd_report.html`, toolpath GIF, uncut 3D renders
Visualisation	Manual ParaView	Embedded PyVista viewport, standalone PyVista GPU window, Three.js WebGL studio (assets inlined for offline use) with heat-source inspector and G-code playback 1×–50×; ParaView still one click away
History output	None	Abaqus-style `*OUTPUT,HISTORY` equivalent: probe points and paths sampled by the solver at true simulated times

12. Workflow and User Experience

A telling detail of continuity: the GUI's progress bar parses *the original script's log lines* (`--- Layer n/N`, `ws=... Tmax=...`, `Inter-pass`), `gui/app.py:6112--6151`.

13. Verification and Validation

14. Lineage: What Came From the Original

14.1 Direct carry-over

1. The θ-scheme 3D transient variational forms — same structure as `aThermal/lThermal`, `downloaded.edp:199--225`.
2. The Goldak double ellipsoid with tanh front/rear split and $f_f = 0.6$, $f_r = 1.4$ — identical algebra.
3. Quiet-element activation with ψ_{active} source masking — the original's central correctness insight, retained everywhere.
4. Progressive x-based node activation inside the weld loop.
5. The inter-pass cooling cadence and the weld / wait / cooldown stage structure of the deposition loop.

Table 9: Workflow comparison.

	Original	**Framework**
Entry point	Edit the file, run one command	`run.py` with ten modes: default (GUI), `--cli`, `--stl`, `--slice`, `--validate`, `--validate-full`, `--generate-only`, `--openfoam`, `--uva`, `--version`
Parameters	40+ constants at the top of the file	Live GUI with 8 presets, units-aware mm/SI conversion, `.waam` project save and load
GUI	None	Tkinter, 12 tabs: Geometry, Process, Material, Mesh/CAD, Physics, Solver, OpenFOAM CFD, Parametric and DOE, 3D Viewport, Script Preview, Ultrasonic, Simulation Log — plus live console, progress parsing, themes and tooltips
Orchestration	Manual	Five stages — slice → mesh → generate → solve → postprocess — into a self-contained `waam_output/<timestamp>/` with a `latest` pointer
Persistence	Rewrite the file	`.waam` projects, `parameters.json` as single source of truth, checkpoint/restart by layer or every N steps
Portability	Hard-coded `D:\freefem++`	`find_freefem_serial/mpi()` plus `FREEFEM_BIN`, `FF_MPIRUN`, `PARAVIEW_BIN` overrides; macOS bundle default; a pseudo-terminal wrapper so FreeFEM's standard output flushes line by line

Table 10: Verification suites present in the framework (none exist upstream).

Suite	**What it checks**
`tests/physics_suite.py` (1,464 lines)	11 groups, 279 assertions: heat-source normalisation, latent enthalpy, VOF, ferrite number, builder markers, static-preset integrity, `.edp` module structure, incremental mechanics, history/restart, structured results, conforming interface
`tests/validate_j2_mechanics.py` (203)	Elastic baseline against J2 plasticity (radial return, TRIP, annealing) on an identical two-layer mesh
`tests/validate_cas_ded_dimeen26.py` (599)	**External benchmark CAS-DEDD-IMEEN26** (Ansys / TU Wien 2026, AlSi10Mg DED): reference targets 0.25 mm max deformation, 4×10^{-2} mm mean, 580 °C process temperature, 50 mm^3/s, 2.5 mm bead radius. Executes a real FreeFEM run, therefore gated behind `--validate-full`
`tests/test_wall_probes_ultrasonic.py` (219)	pytest: deposition -pattern codegen, ultrasonic gating, probe stations, plot generation
`parametric_sweep.py` (324)	Six cases varying current, voltage, wire feed, speed, bead dimensions and layer count, with an incrementally written `sweep_timing_summary.json`

6. Cumulative thermal eigenstrain accumulation $\varepsilon^{\text{th}} \mathrel{+}= \alpha \Delta T$.
7. The boundary-label scheme (1 bottom, 5 top, 2/3/4/6 sides) and the PVD / VTU / `waam_data.csv` output contract.
8. The console progress format — which the GUI now parses.
9. The default geometry: $n_{\text{layers}} = 26$, 200 mm length, 8 mm bead width, 200 A / 24 V / $\eta = 0.8$ / 10 mm/s.

14.2 Evolved or corrected

15. Gap Analysis: What the Framework Still Lacks

1. **No top-level README** comparable to the original's, which documents physics, geometry, ParaView recipes, an error table, extension ideas and a citation. That knowledge now sits inside GUI dialogs and code comments.
2. `core/heat_sources/*.edp` **and** `core/{thermal,mechanical}` **are unwired** — the reference modules are never executed by the pipeline.

Table 11: Places where the framework deliberately departs from the original.

Original	**Framework**
C_p and k both scaled by 10^{-4} in quiet elements	Quench only k, keep full mass: diffusivity collapses, thermal mass does not
No fusion latent heat	Enthalpy-exact piecewise-linear C_p band over $[T_{\mathrm{sol}}, T_{\mathrm{liq}}]$
No radiation or boiling cap	Stefan–Boltzmann plus T_{boil} clip with E_{cap} audit
Eigenstrain may exceed yield → phantom stress	Correct eigenstress subtraction, or full J2 plasticity
`ThBase + ThDep` non-conforming interface	Single conforming mesh
Hard-coded `D:\` output	Portable, timestamped run directories

3. `waam_parametric_sweep_comparison_report.pdf` **is not reproducible** — no code in the repository generates it.
4. **Two millimetre-detection heuristics coexist** — `mesh/units.py` per-key thresholds versus the blanket `>0.05` / `>0.1` test in `run.py`.
5. `test_wall_probes_ultrasonic.py` **is outside** `--validate` — a routine health check misses it unless pytest is run separately.
6. **Multi-material per-layer alloy switching is declared but unimplemented**; only the zonal substrate/deposit split works.
7. **The report is HTML, not PDF**, despite "LaTeX PDF-style" styling.
8. **Licensing** — see Section 16.

16. Licensing and Provenance

Table 12: Licensing of the two works.

	Original	**Framework**
License	CC BY-NC 4.0	GNU GPL v3.0 or later
Holder	Akshansh Mishra (akshansh11)	Pranay1004 and the WAAM_IIST contributors
DOI / URL	`https://doi.org/10.5281/zenodo.20753459`	`https://github.com/Pranay1004/WAAM_IIST`
Commercial use	Not permitted (NC field)	Permitted; derivatives must remain GPL
Files in this repository	`original_git_file/` retained verbatim	`license/LICENSE`, `license/README.md`, `license/NOTICE`

CC BY-NC 4.0 carries a *NonCommercial* field, while the GPL deliberately forbids additional restrictions; the two are therefore not license-compatible in the direction of relicensing upstream code as GPL. The repository does *not* relicense upstream: `original_git_file/` is kept byte-for-byte under CC BY-NC 4.0, only the framework's own source is offered under GPL-3.0-or-later, and a recipient must honour both regimes. The algorithmic ideas themselves (Goldak source, quiet-element activation, progressive torch-triggered birth) are uncopyrightable *methods* and are reimplemented from scratch in `codegen/edp_builder.py`; no upstream source text is copied into the generator. Third-party components keep their own terms (three.js and OrbitControls: MIT; FreeFEM++ and OpenFOAM: GPL, invoked externally and never redistributed).

17. Verdict

Conclusion. The framework is a strict superset of the original: it keeps the original's solver engine (FreeFEM++), Goldak and quiet-element physics, output contract and even its log format, then adds everything a usable simulation platform requires. Its one regression relative to the original is *legibility* — 534 self-contained, heavily commented lines became a 4,992-line generator plus forty Python modules, and the original's exemplary top-level README was not carried forward. The remedy

Table 13: Qualitative scoring of the two code bases.

Dimension	**Orig.**	**Fwk.**	**Comment**
Correctness of the core thermal solve	Good	Stronger	Conforming mesh, energy audit, latent heat, radiation close the original's four physical omissions
Fidelity of the mechanics	Weak	Strong	Elastic estimate versus validated incremental J2
Breadth of physics	Narrow	Broad	+ CFD, UVA, microstructure, hardness
Realism of geometry and toolpath	Poor	Strong	CAD/G-code, five infill patterns
Usability	Weak	Strong	GUI, reports, 3D viewers, project files
Verification	None	Strong	Four suites plus an external benchmark
Documentation	Excellent	Fair	The original's single README was not carried forward
Accessibility of the physics	Excellent	Weak	534 readable lines versus a 4,992-line generator
Dependency weight	Minimal	Heavy	FreeFEM++ only, versus FreeFEM + Gmsh + PyVista + VTK + matplotlib + three.js (+ optional OpenFOAM, pywebview)

suggested by this comparison is modest: restore a single narrative README at the repository root that states the physics, the file map and the validation status in the same voice as the upstream document.

References

[1] A. Mishra. *WAAM-MultiLayer-Simulation*. Zenodo, 2026. `https://doi.org/10.5281/zenodo.20753459`. Licensed CC BY-NC 4.0.

[2] P. K. Pandey and the WAAM_IIST contributors. *WAAM-Framework / WAAM_IIST*, 2026. `https://github.com/Pranay1004/WAAM_IIST`. Licensed GPL-3.0-or-later.

[3] M. Goldak, A. Chakravarti and M. Bibby. A new finite element model for welding heat sources. *Metallurgical Transactions B*, 17B:187–193, 1986.

[4] Progress in Additive Manufacturing, 1:10:11647, 2025 (P48 arc/droplet energy-partition heat source).

[5] CAS-DEDD-IMEEN26 external validation case (Ansys Academic Program / TU Wien, 2026, AlSi10Mg DED).

[6] FreeFEM++ finite-element solver, `https://freefem.org`, version 4.14/4.15.

[7] OpenFOAM `beamWeldFoam`, OpenCFD Ltd., version 2412.